Assessment Book — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12

Questions

Total marks 74

Q1. (4 marks) State the operational definition of the electric field at a point in space using a test charge, and give its SI unit (including an equivalent unit).

Working:

____

____

____

Q2. (1 mark) A point charge q1 = +3.0 μC is located at x = 0.00 m and a point charge q2 = -2.0 μC is located at x = +0.50 m. Using Coulomb's law (k = 8.99 × 10^9 N·m^2/C^2), calculate the magnitude of the electrostatic force on q1 due to q2 and state its direction along the x-axis. Show units in your answer.

- A. Approximately 0.216 N directed to the right (toward +x), because opposite charges attract.
- B. Approximately 0.216 N directed to the left (toward −x), because the product q1·q2 is negative so the force is repulsive.
- C. Approximately 0.0432 N directed to the right (toward +x), from using k·|q1·q2|/r instead of r^2 in Coulomb's law.
- D. Approximately 2.16 N directed to the right (toward +x), from converting microcoulombs incorrectly and overestimating by a factor of 10.

Q3. (3 marks) Using the relation E(r) = (1/4πε0) Q / r^2 r̂, find the electric field vector E at a distance r from an isolated point charge Q. Give the expression for its magnitude with units and state the direction of the field for (a) Q > 0 and (b) Q < 0. (3 marks)

Q4. (6 marks) Use Gauss's law to derive expressions for the electric field magnitude and direction in the following three cases. In each part state the Gaussian surface you choose, the symmetry argument that justifies it, and show the algebraic steps that lead to the final formula (include units for physical constants and clearly state what each symbol means).

(a) An infinite plane with uniform surface charge density σ ($C \cdot m^{-2}$). Give E for points on either side of the plane.

(b) An infinitely long straight wire with uniform linear charge density λ ($C \cdot m^{-1}$). Give E at a distance r from the wire (r measured perpendicular to the wire).

(c) A uniformly charged thin spherical shell of total charge Q (C) and radius R. Give E for (i) $r < R$ (interior points) and (ii) $r > R$ (exterior points), where r is the radial distance from the centre.

Q5. (4 marks) The elementary charge of an electron is $e = 1.60 \times 10^{-19}$ C (negative for electrons). Two neutral insulating objects A and B are rubbed together so that 2.0×10^{12} electrons are transferred from A to B.

(a) Calculate the net electric charge on object A and on object B after the transfer. Give your answers with correct signs and units. (2 marks)

(b) Explain, using the microscopic origin of charge, why (i) the net charge on each object is an integer multiple of e and (ii) the total charge of the isolated two-object system is conserved. Briefly address the common misconception that rubbing 'creates' new charge. (2 marks)

Working:

Q6. (1 mark) An electric dipole consists of two point charges -q at x = -a and +q at x = +a on the x-axis. The dipole moment is therefore p = q(2a) i (pointing from -q to +q). The dipole is placed in a uniform electric field $E = E_0(\cos\theta\, i + \sin\theta\, j)$. Using p = q(2a), find the torque τ on the dipole (give the vector) and state whether a net force acts on the dipole.

- A. $\tau = (2aq E_0 \sin\theta)\, k$ (N·m); net force = 0
- B. $\tau = (2aq E_0 \cos\theta)\, k$ (N·m); net force = 0
- C. $\tau = -(2aq E_0 \sin\theta)\, k$ (N·m); net force = 0
- D. $\tau = (2aq E_0 \sin\theta)\, k$ (N·m); net force = $2qE_0 \cos\theta\, i$ (N)

Q7. (3 marks) Two identical positive point charges, each of charge +Q, are fixed 0.10 m apart. (a) Using Coulomb's law, write an expression for the magnitude of the electrostatic force that one charge exerts on the other and state its direction. (b) Explain why the electric field at the midpoint between the charges is zero, yet each charge still experiences a non-zero force.

Q8. (6 marks) Explain how each of the following everyday phenomena is an example of the discharge of static electricity: (a) a small spark when you take off a synthetic jumper, (b) the shock felt on touching a metal door handle after walking on a carpet, and (c) lightning. In your answer state the physical cause of the charge build-up, describe how the discharge happens, and give at least one everyday-scale numerical anchor (a typical voltage or electric field) to show the difference in scale between the small sparks and lightning.

Q9. (4 marks) Two small insulated spheres are 0.50 m apart. One sphere has charge +6.0 μC and the other has charge +2.0 μC. Calculate the magnitude of the electrostatic force between the spheres and state whether the spheres attract or repel each other. Also give the direction of the force on the +6.0 μC sphere along the line joining them. (Use Coulomb's constant $k = 8.99 \times 10^9\ N{\cdot}m^2/C^2$.)

Working:

Q10. (1 mark) A small metal sphere and a plastic rod are each given a localized +5.0 μC charge at one point on their surface. After they sit undisturbed for a minute in air, where will the added positive charge be located on each object?

- A. On the metal sphere the +5.0 μC has spread over the sphere's outside surface; on the plastic rod the +5.0 μC remains near the point where it was placed.
- B. On the metal sphere the +5.0 μC stays at the original point but is reduced in magnitude; on the plastic rod the +5.0 μC spreads evenly along the rod.
- C. Both the metal sphere and the plastic rod will have the +5.0 μC distributed uniformly throughout their volumes.
- D. Both the metal sphere and the plastic rod will lose most of the +5.0 μC to the surrounding air, so little charge remains on either object.

Q11. (3 marks) Two equal positive point charges +Q are fixed on the x-axis at x = −a and x = +a. Using the principle of superposition for electric fields, state and explain the net electric field at the origin (x = 0).

Q12. (4 marks) A uniform electric field of magnitude 200 N/C passes through a flat square plate of area 0.50 m^2. The electric field makes an angle of 30.0° with the normal to the plate. Calculate the electric flux through the plate. Give your answer with appropriate units and show your working.

Working:

Q13. (1 mark) A small electric dipole is made of charges +q and −q separated by distance a along the z-axis. For points far from the dipole (r >> a) the dipole can be described by its dipole moment p = q a (pointing in +z). Which of the following gives the far-field electric field (magnitude with sign to indicate direction along the radial line from the dipole center) at (i) a point on the +z axis (θ = 0) and (ii) a point in the equatorial plane (θ = 90°)? Include the r-dependence and SI units (N/C).

- A. On axis (θ = 0): E = + (1 / (2πε0))·(p / r^3) N/C; Equator (θ = 90°): E = + (1 / (4πε0))·(p / r^3) N/C
- B. On axis (θ = 0): E = + (1 / (2πε0))·(p / r^3) N/C; Equator (θ = 90°): E = − (1 / (4πε0))·(p / r^3) N/C
- C. On axis (θ = 0): E = − (1 / (2πε0))·(p / r^3) N/C; Equator (θ = 90°): E = − (1 / (4πε0))·(p / r^3) N/C
- D. On axis (θ = 0): E = + (1 / (4πε0))·(p / r^2) N/C; Equator (θ = 90°): E = − (1 / (8πε0))·(p / r^2) N/C

Q14. (3 marks) Two point charges, +2.0 μC and −3.0 μC, are separated by 0.10 m in vacuum. Using Coulomb's law with k = 8.99 × 10^9 N·m^2/C^2, calculate the magnitude of the electrostatic force on the +2.0 μC charge and state its direction (give direction relative to the other charge).

Q15. (4 marks) A) A thin insulating rod of length 0.50 m carries a total charge of 8.0 μC uniformly distributed along its length. Calculate the linear charge density λ and give its SI unit.

B) A thin circular disk of radius 5.0 cm carries a total charge of 1.0×10^{-6} C uniformly distributed over its face. Calculate the surface charge density σ and give its SI unit.

C) A small cube of side 2.0 cm carries a total charge of 2.0 nC distributed uniformly through its volume. Calculate the volume charge density ρ and give its SI unit.

Working:

Q16. (1 mark) Two identical positive point charges Q are fixed a distance 2d apart on a horizontal line. A small positive test charge q is placed exactly at the midpoint (distance d from each Q). Using the superposition principle, what is the net electric force on q?

- A. Zero newtons, because the equal-magnitude repulsive forces from the two Q charges act in opposite directions and cancel.
- B. A repulsive force of magnitude $2kQq/d^2$ to the right, because the forces add algebraically toward the nearer charge.
- C. A repulsive force of magnitude kQq/d^2 to the left, because the left charge dominates by symmetry.
- D. Zero newtons, because the two positive charges neutralize each other and produce no field anywhere between them.

Q17. (3 marks) State the operational definition of the electric field and give its SI unit.

Q18. (4 marks) Two point charges are fixed on a horizontal line: $q_1 = +3.0$ μC is at the origin and $q_2 = -2.0$ μC is at $x = +0.50$ m. Using Coulomb's law, calculate (a) the magnitude of the electrostatic force exerted by q1 on q2, and (b) the direction of this force (state whether it is toward +x or −x). Use $k = 8.99 \times 10^9$ N·m^2·C^−2. (4 marks)

Working:

Q19. (1 mark) A point charge Q = +5.0 μC is placed at the origin. Using $E(r) = (1/4\pi\varepsilon_0)\, Q / r^2\, \hat{r}$, calculate the electric field at the point x = +2.0 m on the x-axis. Give the magnitude with units and state the field direction for the given positive Q and for the case Q = −5.0 μC.

- A. Magnitude $E = 1.12 \times 10^4$ N/C at x = +2.0 m; for Q = +5.0 μC the field points in the +x direction (away from the charge); for Q = −5.0 μC the field points in the −x direction (toward the charge).
- B. Magnitude $E = 2.25 \times 10^4$ N/C at x = +2.0 m; for Q = +5.0 μC the field points in the +x direction; for Q = −5.0 μC the field also points in the +x direction because sign only flips the force on a positive test charge.
- C. Magnitude $E = 1.12 \times 10^4$ N/C at x = +2.0 m; for Q = +5.0 μC the field points in the −x direction (toward the charge); for Q = −5.0 μC the field points in the +x direction (away from the

charge).

- D. Magnitude E = 4.50 × 10^3 N/C at x = +2.0 m; for Q = +5.0 μC the field points in the +x direction; for Q = −5.0 μC the direction is undefined because a negative source charge creates no field sign.

Q20. (4 marks) Use Gauss's law to derive expressions for the magnitude and direction of the electric field for each of the following charge distributions. In each case state the Gaussian surface you choose and give the final answer in terms of the given charge density and epsilon_0.

(a) An infinite uniformly charged plane with surface charge density sigma (C m^-2).
(b) An infinitely long straight wire with linear charge density lambda (C m^-1). Give the field at radial distance r from the wire axis.
(c) A uniformly charged thin spherical shell of radius R with total charge Q. Give the field for r<R (inside) and for r>R (outside).

Working:

____

____

____

Q21. (1 mark) Two identical plastic rods are rubbed with wool and end up with equal and opposite charges. Which statement best explains these observations using the microscopic origin of charge and the principle of charge conservation?

- A. Rubbing creates new elementary charges so each rod gains charge equal to some multiple of the electron charge e (1.60×10^-19 C); the total charge increases but rods show opposite signs because creation happens with opposite sign on each rod.
- B. Rubbing transfers electrons (each with charge −e = −1.60×10^-19 C) from one rod to the other so each rod’s net charge is an integer multiple of e; the total charge of the two-rod system remains constant because no net charge is created or destroyed.
- C. Rubbing rearranges charges so that each rod continuously carries fractional parts of e (e.g., 0.5 e) over macroscopic regions, and the total observed opposite charges sum to a whole number of e only on average.
- D. Rubbing converts some neutral atoms into charged atoms by changing their protons, so the measured charges are not constrained to integer multiples of e and the total charge can change if protons shift between objects.

Q22. (4 marks) An electric dipole consists of charges +q and −q separated by distance 2a. The dipole moment p points from the negative to the positive charge. Let q = 2.0 × 10^−6 C and a = 0.050 m, so p = q(2a). The dipole is placed in a uniform electric field E = 1.0 × 10^4 N C^−1 that points in the +y direction. The dipole moment points in the +x direction. (a) Calculate the torque τ on the dipole (give magnitude, units, and vector direction). (b) State whether there is any net force on the dipole and briefly explain why. [4 marks]

Working:

____

____

____

Q23. (4 marks) Two point charges lie on the x-axis: +8.0 μC at x = 0.00 m (left) and −2.0 μC at x = 0.50 m (right). (a) Find the magnitude and direction of the electric field at the midpoint (x = 0.25 m). (b) A test charge +1.0 μC is placed at the midpoint. Find the magnitude and direction of the electrostatic force on the test charge. Use $k = 8.99 \times 10^9$ N·m^2/C^2.

Working:

Q24. (4 marks) Air breaks down and produces a spark when the electric field exceeds about 3.0×10^6 V/m. (a) Calculate the voltage difference needed to produce a spark across a 1.0 cm gap (typical spark between synthetic clothes). (b) Calculate the voltage difference needed to produce a spark across a 2.0 mm gap (typical small shock when touching a metal handle). (c) Based on your calculations, state which spark requires the larger voltage and give one everyday reason why one type of spark is more common than the other. (Use 3.0×10^6 V/m for the breakdown strength of air. Give answers with correct units.)

Working:

Answer Key

Q1.

Answer: The electric field E at a point is defined operationally as the force F experienced per unit positive test charge q placed at that point: $E = F/q$ (using a positive, negligibly small test charge so it does not disturb the source charges). Its SI unit is newtons per coulomb (N C^-1), which is equivalent to volts per metre (V m^-1).

Working: 1. Definition: Place a small positive test charge q at the point; measure the force F on it. By definition $E = F/q$. This specifies the field as force per unit positive test charge and implies the test charge should be negligibly small so it does not disturb the source charge distribution.

2. Units: Force F has units N (newtons) and charge q has units C (coulombs), so E has units N/C.

3. Equivalence to V/m: 1 N = 1 kg·m·s^-2, and 1 volt V = 1 J/C = 1 (N·m)/C, so N/C = (N·m)/(C·m) = V/m. Therefore N C^-1 is equivalent to V m^-1.

Rubric — Correct operational definition (force per unit positive test charge, test charge negligible) and correct SI unit with equivalent unit.

Full credit (4 marks): Gives the operational definition $E = F/q$ with explicit statement that q is a positive, negligibly small test charge (so the source is not disturbed), and states the SI unit as N C^-1 and its equivalence V m^-1.

Partial credit (2 marks): Gives $E = F/q$ but omits that the test charge should be positive or negligibly small, or gives only one correct unit (either N C^-1 or V m^-1) without noting the equivalence.

No credit (0 marks): Definition incorrect or missing, and unit incorrect or missing.

Q2.

Correct: A. Approximately 0.216 N directed to the right (toward +x), because opposite charges attract.

A. Correct: Convert charges to coulombs $q_1 = 3.0\times10^{-6}$ C, $q_2 = -2.0\times10^{-6}$ C. Magnitude = $k\cdot|q_1\cdot q_2|/r^2 = (8.99\times10^9)(6.0\times10^{-12})/(0.50^2) = (8.99\times10^9)(2.4\times10^{-11})$ 0.216 N. Charges are opposite so they attract; q1 feels a force toward q2, i.e. to the right (+x). Units N are included.

B. Wrong direction: While the magnitude 0.216 N is correct, the force is toward q2 (attraction), so it acts to the right. Saying left confuses sign of q1·q2 with the physical direction of attraction for opposite charges.

C. Wrong magnitude: This option gives a smaller value because it uses r (0.50 m) instead of r^2 in the denominator. Coulomb's law requires division by r^2, so the result is too small by a factor of r (0.50 m).

D. Wrong magnitude: This is too large by a factor of 10. It reflects an arithmetic error when converting microcoulombs to coulombs or a miscalculation in the exponent; proper conversion and calculation give about 0.216 N, not 2.16 N.

Q3.

Answer: $E(r) = (1 / (4\pi\varepsilon_0)) \cdot (Q / r^2)\,\hat{r}$. Magnitude: $|E| = (1 / (4\pi\varepsilon_0)) \cdot |Q| / r^2$, in units of newtons per coulomb (N C^-1) or volts per metre (V m^-1). Direction: (a) For $Q > 0$ the field points radially outward from the charge ($\hat{r}$). (b) For $Q < 0$ the field points radially inward toward the charge ($-\hat{r}$).

Rubric — Correct expression for E with units and correct direction for positive and negative Q

Full (3 marks) (3 marks): Gives the vector expression $E(r) = (1/4\pi\varepsilon_0)\cdot(Q / r^2)\,\hat{r}$, states the magnitude $|E| = (1/4\pi\varepsilon_0)\cdot|Q|/r^2$ with correct units (N C^-1 or V m^-1), and correctly states that for $Q>0$ the field is radially outward ($\hat{r}$) and for $Q<0$ it is radially inward ($-\hat{r}$).

Partial (1–2 marks) (2 marks): Gives the correct formula either as a vector or magnitude but omits units or gives incorrect/ambiguous direction for one sign of Q, or gives magnitude without absolute value so sign/direction treatment is unclear. Award 2 marks for correct formula and units but one directional statement missing/incorrect; award 1 mark for a correct magnitude only (with or without units).

Minimal / Incorrect (0 marks) (0 marks): Expression and/or direction are incorrect or missing; units absent and no clear statement about direction for Q>0 and Q<0.

Q4.

Answer: (a) Infinite plane (surface charge density σ, units $C \cdot m^{-2}$):

Symmetry: field is perpendicular to plane and same magnitude at equal distances on both sides. Choose a cylindrical Gaussian pillbox of cross-sectional area A (m^2) whose flat faces are parallel to the plane and extend equally to both sides.

Gauss's law: E·dA = Q_enclosed/ε0.

Flux through curved side = 0 (E perpendicular to curved side). Flux through two flat faces = E·A + E·A = 2EA.

Enclosed charge = σA.

So 2EA = σA/ε0 E = σ/(2ε0). Direction: normal to the plane, away from plane if σ>0; toward plane if σ<0. Units: σ ($C \cdot m^{-2}$)/ε0 ($C^2 \cdot N^{-1} \cdot m^{-2}$) gives E in $N \cdot C^{-1}$ (or $V \cdot m^{-1}$).

(b) Infinite line (linear charge density λ, units $C \cdot m^{-1}$):

Symmetry: field is radial (perpendicular to wire) and depends only on radial distance r. Choose a coaxial cylindrical Gaussian surface of radius r and length L (m).

Gauss's law: E·dA = Q_enclosed/ε0.

Field is constant on curved surface and perpendicular to it, zero flux through end caps (field parallel to end caps). Flux = E(2πrL).

Enclosed charge = λL.

So E(2πrL) = λL/ε0 E = λ/(2π ε0 r). Direction: radially outward for λ>0, inward for λ<0. Units: $C \cdot m^{-1}$/(ε0·m) $N \cdot C^{-1}$.

(c) Thin spherical shell (total charge Q, radius R):

Symmetry: field is radial and depends only on r. Use a concentric spherical Gaussian surface of radius r and surface area $4\pi r^2$.

Gauss's law: E·dA = Q_enclosed/ε0.

(i) For r < R (interior): Gaussian surface encloses no charge (all charge on the shell). Q_enclosed = 0 E($4\pi r^2$) = 0 E = 0. Direction: zero field (no preferred direction).

(ii) For r > R (exterior): Gaussian surface encloses total charge Q. So E($4\pi r^2$) = Q/ε0 E = Q/(4π ε0 r^2). Direction: radial (away from centre if Q>0). This equals the field of a point charge Q at the centre. Units: C/(ε0·m^2) $N \cdot C^{-1}$.

In all parts ε0 is the vacuum permittivity (ε0 8.85×10^{-12} $C^2 \cdot N^{-1} \cdot m^{-2}$).

Working: (a) Infinite plane:

1. Choose pillbox with area A. By symmetry E normal to plane and same magnitude on both faces.
2. Flux = E·A (top) + E·A (bottom) = 2EA.
3. Q_enclosed = σA.
4. Gauss: 2EA = σA/ε0 cancel A E = σ/(2ε0). Units $N \cdot C^{-1}$.

(b) Infinite line:

1. Choose coaxial cylinder radius r length L. By symmetry E radial and constant on curved surface.

2. Flux through curved surface = E·(2πrL). End caps contribute zero.

3. Q_enclosed = λL.

4. Gauss: E(2πrL) = λL/ε0 cancel L E = λ/(2π ε0 r). Units $N \cdot C^{-1}$.

(c) Spherical shell:

1. Choose concentric sphere radius r. By symmetry E radial and constant over sphere.

2. Flux = $E \cdot (4\pi r^2)$.

3. (i) r < R: Q_enclosed = 0 $E(4\pi r^2) = 0$ E = 0.

4. (ii) r > R: Q_enclosed = Q $E(4\pi r^2) = Q/\varepsilon_0$ $E = Q/(4\pi \varepsilon_0 r^2)$. Units $N \cdot C^{-1}$.

5. Note: ε0 $8.85\times10^{-12}\ C^2 \cdot N^{-1} \cdot m^{-2}$.

Rubric — Correct, complete derivations using Gauss's law for each of the three charge distributions, with stated Gaussian surfaces, symmetry arguments, algebraic steps, and correct final expressions with units and directions.

Full credit (6 marks): All three cases correctly derived (plane: E = σ/(2ε0) with pillbox and 2EA = σA/ε0; line: E = λ/(2π ε0 r) with coaxial cylinder and E(2πrL)=λL/ε0; sphere: interior E=0 and exterior $E=Q/(4\pi \varepsilon_0 r^2)$ using spherical surface), symmetry and Gaussian surfaces stated, algebra shown, units and direction given.

Partial credit (3 marks): One or two cases correctly derived with required steps and units, or all three attempted but with a significant algebraic or reasoning error in one case (e.g. wrong factor 2 for plane or missing 2π factor for wire). Symmetry or Gaussian surface may be briefly stated but not fully justified.

No credit (0 marks): Derivations missing or incorrect for all three cases, Gaussian surfaces not chosen or symmetry arguments absent, or final expressions incorrect and unsupported by algebra.

Q5.

Answer: (a) Charge on A: -2.0 × 10^12 electrons removed loss of negative charge means A becomes positive: Q_A = + (2.0 × 10^12)(1.60 × 10^-19 C) = +3.20 × 10^-7 C. Charge on B: gained 2.0 × 10^12 electrons Q_B = - (2.0 × 10^12)(1.60 × 10^-19 C) = -3.20 × 10^-7 C.

(b) (i) Each electron has charge -e and each proton has charge +e, so any net free charge is a count of excess or deficit electrons (or protons) and thus must be an integer multiple of the elementary charge e. In this example the charges are ±(2.0 × 10^12)e, giving the values above. (ii) Charge is conserved because rubbing only moves discrete charge carriers (electrons) from one object to the other; no net charge is created or destroyed in the isolated two-object system, so the algebraic sum Q_A + Q_B remains zero (+3.20 × 10^-7 C + (−3.20 × 10^-7 C) = 0). The common misconception is wrong: rubbing does not create charge but transfers existing electrons between objects.

Working: (a) Number of electrons transferred: N = 2.0 × 10^12 (dimensionless).

Charge per electron: e = 1.60 × 10^-19 C (magnitude).

Charge moved (magnitude) = N × e = (2.0 × 10^12)(1.60 × 10^-19 C) = 3.20 × 10^-7 C.

Object A loses N electrons, so its net charge Q_A = + (N e) = +3.20 × 10^-7 C.

Object B gains N electrons, so its net charge Q_B = - (N e) = -3.20 × 10^-7 C.

Check conservation: Q_total = Q_A + Q_B = +3.20 × 10^-7 C + (−3.20 × 10^-7 C) = 0 C.

(b) Microscopic origin: electrons (charge −e) and protons (charge +e) are discrete. Any change in net charge equals an integer number of these charge carriers multiplied by e, so charges are quantised. Rubbing moves electrons from one body to the other; it does not create or destroy electron/proton pairs in this isolated system, so total charge is conserved.

Rubric — Correct numerical charges with units; explanation that charge is quantised in units of e and that total charge is conserved by transfer of electrons, addressing the rubbing misconception.

Full credit (4 marks): Calculations correct with signs and units: $Q_A = +3.20 \times 10^{-7}$ C and $Q_B = -3.20 \times 10^{-7}$ C (2 marks). Explanation correctly states that net charge is an integer multiple of e because electrons/protons are discrete carriers, and that rubbing transfers electrons so total charge of the isolated system is conserved; explicitly refutes the idea that rubbing creates net charge (2 marks).

Partial credit (2 marks): Either the numerical part is correct but explanation incomplete or partly incorrect (e.g. states charge is discrete but does not mention electrons/protons or conservation), or explanation correct but numerical values have a factor or sign error (numbers within a factor of 10 or wrong sign) with correct units. Award 2 marks.

No credit (0 marks): Numerical answers incorrect with no useful working, and explanation fails to state quantisation from discrete charge carriers or fails to address conservation and the rubbing misconception.

Q6.

Correct: A. $\tau = (2aq E_0 \sin\theta)\, k$ (N·m); net force = 0

A. Correct: compute $\tau = p \times E = [2aq\, i] \times [E_0(\cos\theta\, i + \sin\theta\, j)] = 2aq E_0 \sin\theta\, (i \times j) = 2aq E_0 \sin\theta\, k$. In a uniform field the forces on +q and −q are equal and opposite so the net force is zero. Units: p (C·m) × E (N/C) N·m.

B. Misconception: this option uses cosθ instead of sinθ, which arises from incorrectly projecting E along i for the cross product; i×i = 0, the cross term producing torque is from the i×j component which gives sinθ.

C. Sign error: this option has a negative sign. That mistake comes from reversing the order of the cross product or the sign convention for p; with p = +2aq i and the given E the cross product gives a positive k-component, not negative.

D. Force-error: this option gives the correct torque magnitude but claims a net force. In a uniform field the forces on +q and −q are equal in magnitude and opposite, so they cancel — there is no net force. The listed nonzero force confuses dipoles in nonuniform fields with uniform-field behavior.

Q7.

Answer: (a) Coulomb's law gives the magnitude of the force between two point charges as $F = k\cdot|Q_1\cdot Q_2|/r^2$. For these charges $Q_1 = Q_2 = Q$ and $r = 0.10$ m, so $F = k\cdot Q^2/(0.10\ \text{m})^2$. Using $k = 8.99\times10^9\ \text{N·m}^2/\text{C}^2$, the magnitude is $F = (8.99\times10^9\ \text{N·m}^2/\text{C}^2)\cdot Q^2/(0.01\ \text{m}^2) = 8.99\times10^{11}\cdot Q^2\ \text{N/C}^2$ (note Q in coulombs gives F in newtons). The direction is along the line joining the charges: they repel, so each charge is pushed directly away from the other.

(b) At the midpoint, the electric field contributions from the two identical positive charges have equal magnitude and opposite directions, so they cancel and the net electric field is zero. However, the force on a given charge is produced by the electric field due to the other charge at the location of that given charge; a charge does not exert a force on itself. Therefore each charge feels the non-zero repulsive force from the other charge (given by the expression in part (a)) even though the field at the midpoint between them is zero.

Rubric — Correct use of Coulomb's law, correct direction of force, and clear explanation distinguishing zero field at midpoint from non-zero force on each charge.

Full credit (3 marks): Gives the correct Coulomb expression with r = 0.10 m substituted ($F = k\cdot Q^2/(0.10\ \text{m})^2$) and units indicated, states the correct direction (repulsive, along the line between charges), and clearly explains that fields from the two charges cancel at the midpoint while each charge experiences a force due to the other charge (explicitly noting a charge does not act on itself).

Partial credit (2 marks): Gives the correct Coulomb formula but omits substituting r or units, or gives the correct direction but the explanation about field cancellation versus force is incomplete or only partially correct (e.g. mentions cancellation but does not state why each charge still feels a force).

Minimal credit (1 marks): States qualitatively that the charges repel and that fields cancel at the midpoint or that each charge feels a force from the other, but either gives no Coulomb expression or gives an incorrect/contradictory explanation.

Q8.

Answer: All three phenomena are discharges of static electric charge caused by a build-up of charge (usually excess electrons or a net positive deficit of electrons) and a later rapid movement of charge to neutralise a potential difference.

(a) Synthetic jumper spark: When you pull off or rub a synthetic jumper, friction causes transfer of electrons between the fibres and your body (triboelectric effect). The body or jumper becomes charged, producing a potential difference of order 1 000–3 000 V relative to surrounding objects. If the potential difference is large enough across a small gap, the air breaks down and a tiny spark (a brief flow of electrons) jumps, equalising the charge. The spark is short (millimetres) and produces a small shock and sometimes a visible snap and smell of ozone.

(b) Shock from a metal door handle after walking on carpet: Walking across carpet causes repeated contact and separation between shoe soles and carpet fibres, transferring charge so your body can accumulate a net charge (often negative). A metal handle is a good conductor and provides a path to ground. When you reach and touch the handle, the potential difference between your charged body and the handle causes electrons to move quickly through the small contact gap — a discharge you feel as a small shock. Typical voltages producing felt shocks are around 1 000–3 000 V; the actual charge is small, so the energy is low but enough to be felt.

(c) Lightning: In storm clouds, collisions between ice particles and water droplets separate charges so large regions of a cloud become strongly charged (positive and negative regions). The potential difference between cloud and ground (or between cloud regions) can reach tens to hundreds of millions of volts (10^7–10^9 V). When the electric field exceeds air breakdown (roughly 3 × 10^6 V m^-1), a conductive channel forms and a huge current flows — the lightning stroke — rapidly neutralising charge over kilometres and releasing large energy, heat, light, and thunder.

Common mechanism: In all cases charge separation creates a potential difference; when the local electric field exceeds the breakdown threshold of air (or a conductive path is provided), rapid movement of charge (a discharge) occurs. The main differences are scale (small sparks: ~10^3 V, short gaps; lightning: ~10^7–10^9 V, long channels) and energy delivered (small shocks have very low energy; lightning releases very large energy).

Rubric — Correct, linked explanations that show how each example is a static discharge, including the cause of charge build-up, the discharge process, and at least one numerical anchor.

Full credit (6 marks) (6 marks): Identifies all three examples and for each explains (1) how charge is built up (triboelectric collisions or charge separation in clouds), (2) how discharge occurs (air breakdown or conduction to ground) and (3) gives at least one correct everyday numerical anchor for scale (e.g. ~1 000–3 000 V for jumper/door-handle sparks, ~10^7–10^9 V for lightning). Explanations link the cause, the resulting potential difference/electric field, and the spark/shock/lightning as the rapid neutralisation of charge.

Partial credit (3 marks) (3 marks): Identifies the examples and gives a qualitatively correct explanation for most parts (charge build-up and that a discharge neutralises it) but omits or gives incorrect/absent numerical anchors or fails to fully link cause and discharge for one or more examples.

Minimal/no credit (0 marks) (0 marks): Only lists phenomena without explanation, gives incorrect mechanisms (e.g. chemical causes rather than charge transfer) or presents entirely wrong numerical values; does not show understanding of discharge as rapid neutralisation of static charge.

Q9.

Answer: Magnitude: 4.32 × 10^-1 N (approximately 0.432 N). The two spheres repel each other because both charges are positive. The force on the +6.0 μC sphere is directed away from the +2.0 μC sphere along the line joining them.

Working: Given q1 = +6.0 μC = 6.0 × 10^-6 C, q2 = +2.0 μC = 2.0 × 10^-6 C, r = 0.50 m, k = 8.99 × 10^9 N·m^2/C^2.

Coulomb's law: F = k |q1 q2| / r^2.

Compute product of charges: q1 q2 = (6.0 × 10^-6 C)(2.0 × 10^-6 C) = 12.0 × 10^-12 C^2 = 1.20 × 10^-11 C^2.

Compute r^2 = (0.50 m)^2 = 0.25 m^2.

Compute F = (8.99 × 10^9 N·m^2/C^2) × (1.20 × 10^-11 C^2) / (0.25 m^2)

= (8.99 × 10^9 × 1.20 × 10^-11) / 0.25 N

= (1.0788 × 10^-1) / 0.25 N

= 4.3152 × 10^-1 N 4.32 × 10^-1 N.

Since both charges are positive, like charges repel. Therefore the force is repulsive. The force on the +6.0 μC sphere is directed away from the +2.0 μC sphere along the line joining them.

Rubric — Correct numerical magnitude of force and correct identification of attraction/repulsion with direction for the +6.0 μC sphere.

Full marks (4 marks): Gives the correct magnitude (4.32 × 10^-1 N) with units and shows correct use of Coulomb's law, and states that the spheres repel (both positive) with the force on +6.0 μC directed away from the +2.0 μC sphere along the line joining them.

Partial credit (2 marks): Shows correct method (Coulomb's law) and either obtains the correct magnitude but gives the wrong interaction/direction, OR correctly identifies that they repel/attract and direction but numerical magnitude is significantly incorrect due to a clear arithmetic error; units should be present. (Award 2 marks.)

No credit (0 marks): Gives an incorrect magnitude and an incorrect statement about attraction/repulsion and direction, or shows no working and no units.

Q10.

Correct: A. On the metal sphere the +5.0 μC has spread over the sphere's outside surface; on the plastic rod the +5.0 μC remains near the point where it was placed.

A. Correct: In a conductor (metal sphere) free electrons move and redistribute so excess charge resides on the outer surface; in an insulator (plastic rod) charges are not free to move, so the added charge stays localized near where it was placed. This matches the behavior of conductors and insulators for the given 5.0 μC example.

B. Mistake: Reverses conductor/insulator behavior. Conductors do not keep excess charge at a single point; they redistribute. Insulators typically do not allow charge to spread evenly along the object, so this option reflects a misunderstanding of charge mobility.

C. Mistake: Treats both materials as if charge moves freely through volume. In conductors excess charge moves to the surface, not throughout the volume; in insulators charge usually stays where placed rather than becoming uniform in the volume.

D. Mistake: Assumes charge is quickly lost to air in both cases. While some charge can leak to air over time, the typical and immediate behavior is that conductors redistribute charge on their surface and insulators keep charge localized; this option confuses slow leakage with the fundamental conductor/insulator difference.

Q11.

Answer: The net electric field at the origin is zero. Each charge produces an electric field at the origin of magnitude E = (1/(4πε0))·(Q/a^2). The field from the charge at x = −a points in the −x direction (away from the positive charge) and the field from the charge at x = +a points in the +x direction. These two fields have equal magnitude and opposite directions, so by vector superposition they cancel, giving a resultant electric field of zero at the origin.

Rubric — Correct application of superposition to find the resultant electric field at the origin and clear vector reasoning.

Full marks (3 marks): States that the net field is zero; gives magnitude of each field $E = (1/(4\pi\varepsilon_0))\cdot(Q/a^2)$ and correctly identifies their directions (away from each +Q); explicitly states that equal magnitudes and opposite directions cancel by vector superposition.

Partial credit (2 marks): States that the fields cancel and concludes net field is zero but omits the expression for the magnitude of each field or gives only a verbal argument without explicitly citing equal magnitudes and opposite directions as vector cancellation.

Minimal or no credit (0 marks): Gives an incorrect result (not zero) or only describes one charge's field at the origin without using superposition/vector cancellation.

Q12.

Answer: The electric flux through the plate is $\Phi = E A \cos\theta = (200\ \text{N/C})(0.50\ \text{m}^2) \cos 30.0° = 86.6\ \text{N·m}^2/\text{C}$ (to three significant figures).

Working: Step 1 — Write the formula for electric flux through a flat surface:

$\Phi = E \cdot A \cdot \cos\theta$, where

E = magnitude of the electric field (N/C),

A = area of the surface (m^2),

θ = angle between the field and the surface normal (degrees).

Step 2 — Substitute the given values (carry units):

$E = 200$ N/C, $A = 0.50\ \text{m}^2$, $\theta = 30.0°$.

$\Phi = (200\ \text{N/C}) \times (0.50\ \text{m}^2) \times \cos(30.0°)$

Step 3 — Evaluate cos 30.0° = 3/2 0.8660254, and multiply (carry units):

$\Phi = 200 \times 0.50 \times 0.8660254\ \text{N·m}^2/\text{C}$

$\Phi = 100 \times 0.8660254\ \text{N·m}^2/\text{C}$

$\Phi = 86.60254\ \text{N·m}^2/\text{C}$

Step 4 — Quote to appropriate significant figures (given data to three sig figs for area and four for field, report to three sig figs):

Φ 86.6 $\text{N·m}^2/\text{C}$.

Rubric — Correct use of $\Phi = E A \cos\theta$, correct substitution, calculation, and units

Full marks (4 marks): Uses $\Phi = E A \cos\theta$, substitutes $E = 200$ N/C, $A = 0.50\ \text{m}^2$, $\theta = 30.0°$, evaluates cos30 correctly and computes Φ 86.6 $\text{N·m}^2/\text{C}$ with units shown.

Partial credit (2 marks): Shows correct formula $\Phi = E A \cos\theta$ and substitution but has a numerical error in the multiplication or quotes the value with incorrect significant figures or missing decimal precision; units may be present or slightly incorrect (award when method is correct but final number is wrong).

No credit (0 marks): Fails to use $\Phi = E A \cos\theta$ (for example uses $\sin\theta$ or omits the angle), or gives an answer with no working and incorrect units, or shows no understanding of flux through an area.

Q13.

Correct: B. On axis ($\theta = 0$): $E = +\,(1 / (2\pi\epsilon_0))\cdot(p / r^3)$ N/C; Equator ($\theta = 90°$): $E = -\,(1 / (4\pi\epsilon_0))\cdot(p / r^3)$ N/C

A. Incorrect: the equatorial sign is wrong. A positive dipole moment along +z produces a field on the equatorial plane pointing toward the dipole (negative radial component), not away; also the equatorial coefficient should be $1/(4\pi\epsilon_0)$ but with a negative sign.

B. Correct: for r >> a an ideal dipole field falls off as $1/r^3$. Along the +z axis the field points away from the dipole and has magnitude $(1/(2\pi\epsilon_0))\cdot(p/r^3)$ N/C; on the equatorial plane the field points toward the dipole and has magnitude $(1/(4\pi\epsilon_0))\cdot(p/r^3)$ N/C, so the equatorial component carries a negative sign. Units N/C follow from p (C·m) divided by ϵ_0 ($C^2/(N\cdot m^2)$) and r^3 (m^3).

C. Incorrect: both signs are wrong for a dipole with p in +z: on axis above the dipole the field should point away (positive), not toward (negative). The equatorial sign alone is correct but the axis sign is incorrect.

D. Incorrect: the radial dependence is wrong. Far-field dipole terms scale as $1/r^3$, not $1/r^2$, and the numerical coefficients are incorrect for the dipole far-field; these expressions look like monopole-like $1/r^2$ terms, which do not apply to a neutral dipole.

Q14.

Answer: Magnitude: $F = k |q_1 q_2| / r^2 = (8.99 \times 10^9\ N\cdot m^2/C^2) \times (2.0 \times 10^{-6}\ C \times 3.0 \times 10^{-6}\ C) / (0.10\ m)^2 = 5.39 \times 10^{-2}$ N. Direction: the force on the +2.0 μC charge is toward the −3.0 μC charge (attractive).

Working: Step 1: Write Coulomb's law for magnitude: $F = k |q_1 q_2| / r^2$.

Step 2: Substitute values with units: $q_1 = 2.0 \times 10^{-6}$ C, $q_2 = -3.0 \times 10^{-6}$ C, r = 0.10 m, $k = 8.99 \times 10^9\ N\cdot m^2/C^2$.

Step 3: Compute product of magnitudes of charges: $|q_1 q_2| = (2.0 \times 10^{-6}\ C)(3.0 \times 10^{-6}\ C) = 6.0 \times 10^{-12}\ C^2$.

Step 4: Compute $r^2 = (0.10\ m)^2 = 0.0100\ m^2$.

Step 5: Compute force: $F = (8.99 \times 10^9\ N\cdot m^2/C^2) \times (6.0 \times 10^{-12}\ C^2) / (0.0100\ m^2) = (8.99 \times 10^9 \times 6.0 \times 10^{-12} / 0.0100)$ N = 0.05394 N 5.39×10^{-2} N.

Step 6: Determine direction: charges are opposite in sign, so the +2.0 μC charge is attracted toward the −3.0 μC charge; therefore the force on the +2.0 μC is directed toward the other charge.

Rubric — Correct application of Coulomb's law to compute force magnitude with units and correct physical direction (attraction/repulsion).

Full marks (3 marks): Gives the correct magnitude to two significant figures or better with correct unit (5.39×10^{-2} N) and explicitly states the force on the +2.0 μC charge is toward the −3.0 μC charge (attractive). Includes a correct substitution or equivalent calculation.

Partial credit (1 marks): Gives a correct numerical magnitude with unit but either omits the direction or gives an unclear direction, OR states the correct qualitative direction (attractive toward the negative charge) but the magnitude is incorrect by more than reasonable rounding (no correct working shown).

No credit (0 marks): Numeric result and direction both missing or both incorrect, or answer shows no correct use of Coulomb's law.

Q15.

Answer: A) $\lambda = Q / L = (8.0 \times 10^{-6}\ C) / (0.50\ m) = 1.6 \times 10^{-5}\ C\ m^{-1}$.

B) Area of disk $A = \pi r^2 = \pi (0.050\ m)^2 = 7.85398 \times 10^{-3}\ m^2$. $\sigma = Q / A = (1.0 \times 10^{-6}\ C) / (7.85398 \times 10^{-3}\ m^2) = 1.27 \times 10^{-4}\ C\ m^{-2}$.

C) Volume of cube $V = (0.020\ m)^3 = 8.00 \times 10^{-6}\ m^3$. $\rho = Q / V = (2.0 \times 10^{-9}\ C) / (8.00 \times 10^{-6}\ m^3) = 2.50 \times 10^{-4}\ C\ m^{-3}$.

Working: A)

Step 1: Write definition $\lambda = dQ / dl$ Q / L for uniform line charge.

Step 2: Substitute $Q = 8.0 \times 10^{-6}$ C, L = 0.50 m.

λ = (8.0 × 10^−6 C) / (0.50 m) = 1.6 × 10^−5 C m^−1.
Unit: C m^−1 (or C/m).

B)
Step 1: Write definition σ = dQ / dA Q / A for uniform surface charge.
Step 2: Radius r = 5.0 cm = 0.050 m. Area A = π r^2 = π (0.050 m)^2 = π × 2.5 × 10^−3 m^2 = 7.85398 × 10^−3 m^2.
Step 3: Substitute Q = 1.0 × 10^−6 C.
σ = (1.0 × 10^−6 C) / (7.85398 × 10^−3 m^2) = 1.2726 × 10^−4 C m^−2 1.27 × 10^−4 C m^−2.
Unit: C m^−2 (or C/m^2).

C)
Step 1: Write definition ρ = dQ / dV Q / V for uniform volume charge.
Step 2: Side a = 2.0 cm = 0.020 m. Volume V = a^3 = (0.020 m)^3 = 8.00 × 10^−6 m^3.
Step 3: Substitute Q = 2.0 nC = 2.0 × 10^−9 C.
ρ = (2.0 × 10^−9 C) / (8.00 × 10^−6 m^3) = 2.50 × 10^−4 C m^−3.
Unit: C m^−3 (or C/m^3).
Rubric — Correct use of definitions and correct numerical values with appropriate SI units for λ, σ, and ρ.
Full marks (4 marks): All three densities calculated correctly using the correct formulas (λ = Q/L, σ = Q/A, ρ = Q/V), with each numerical value correct to appropriate significant figures and each accompanied by the correct SI unit (C m^−1, C m^−2, C m^−3).
Partial credit (2 marks): Two of the three densities are calculated correctly with correct units, or all three numerical values are correct but one or more units are missing or incorrect.
No credit (0 marks): Fewer than two correct numerical densities and/or major formula errors (wrong use of Q/A, Q/L, Q/V) and units incorrect or absent for the correct parts.

Q16.

Correct: A. Zero newtons, because the equal-magnitude repulsive forces from the two Q charges act in opposite directions and cancel.

A. Correct. Each Q exerts a repulsive Coulomb force of magnitude kQq/d^2 on q; the two forces are equal in magnitude and opposite in direction, so their vector sum is zero newtons.
B. Incorrect: this treats the magnitudes as if they simply add in the same direction. Here the forces are opposite, so they subtract and do not give 2kQq/d^2 to one side.
C. Incorrect: symmetry means neither side dominates. This option confuses choosing a direction with the actual symmetry; the magnitudes are equal so the net is not kQq/d^2 to one side.
D. Incorrect: although the net force is zero at the midpoint, the phrase 'neutralize each other' is misleading physically—charges do not cancel charge, they produce fields that vector-sum to zero at that point. The correct reason is vector cancellation of forces, not literal neutralization.

Q17.

Answer: The electric field at a point is defined operationally as the force experienced per unit positive test charge placed at that point: E = F/q_positive. It is a vector field (direction is the force direction on a positive test charge) and is independent of the magnitude of the test charge provided the test charge is negligibly small so as not to disturb the source charges. The SI unit of electric field is newtons per coulomb (N C^-1), which is equivalent to volts per metre (V m^-1).

Rubric — Correct operational definition of electric field and correct SI unit given.
Full credit (3 marks): Gives the operational definition exactly as 'force per unit positive test charge' (E = F/q_positive), mentions it is a vector and independent of the test charge if the test charge is negligible, and states

the SI unit as N C^-1 (and optionally V m^-1).

Partial credit (2 marks): Gives the operational definition as force per unit charge but omits specifying 'positive' test charge or the requirement that the test charge be negligibly small, or gives the correct definition but omits the SI unit.

Minimal credit (1 marks): Mentions either 'force per unit (test) charge' or states the correct SI unit N C^-1 (or V m^-1), but does not provide the full operational definition (missing vector nature or relation E = F/q_positive).

Q18.

Answer: Magnitude = 0.216 N. Direction = toward the −x direction (toward q1), because the charges are opposite and the force is attractive.

Working: Step 1: Convert charges to SI units.
q1 = +3.0 μC = +3.0 × 10^−6 C.
q2 = −2.0 μC = −2.0 × 10^−6 C.
Separation r = 0.50 m.

Step 2: Write Coulomb's law for magnitude of the force between two point charges:
F = k · |q1 q2| / r^2.

Step 3: Substitute values with units:
|q1 q2| = |(+3.0 × 10^−6 C) · (−2.0 × 10^−6 C)| = 6.0 × 10^−12 C^2.
r^2 = (0.50 m)^2 = 0.25 m^2.

F = (8.99 × 10^9 N·m^2·C^−2) · (6.0 × 10^−12 C^2) / (0.25 m^2).

Step 4: Calculate numerator: 8.99 × 10^9 × 6.0 × 10^−12 = 53.94 × 10^−3 N·m^2 / m^2 = 0.05394 N.
Now divide by 0.25: F = 0.05394 N / 0.25 = 0.21576 N.

Step 5: Report with appropriate significant figures (given data to 2–3 s.f.):
F = 0.216 N (to three significant figures).

Step 6: Determine direction. q1 is positive and q2 is negative, so the force is attractive: q2 is pulled toward q1 which is at the origin (x = 0). q2 is at x = +0.50 m, so the force on q2 is toward −x.

Final answer: Magnitude = 0.216 N; Direction = toward −x (toward q1).
Rubric — Correct calculation of force magnitude using Coulomb's law, with units, and correct statement of direction.

Full credit (4 marks): Correct magnitude (0.216 N) with units shown and correct direction stated as toward −x (attractive toward q1). Arithmetic and units carried through clearly.

Partial credit (2 marks): Correct use of Coulomb's law and correct setup but numerical answer has a clear arithmetic or rounding error (result differs significantly but work shows correct method) or units missing; direction may be correctly stated or omitted. Shows correct intermediate steps (e.g. substitution and r^2) but final value inaccurate.

No credit (0 marks): Wrong method (not using Coulomb's law), incorrect or missing intermediate steps, or answer gives both wrong magnitude and wrong direction.

Q19.

Correct: A. Magnitude E = 1.12 × 10^4 N/C at x = +2.0 m; for Q = +5.0 μC the field points in the +x direction (away from the charge); for Q = −5.0 μC the field points in the −x direction (toward the charge).

A. Correct: compute k = 1/(4πε0)   8.99×10^9 N·m^2/C^2, then E = k|Q|/r^2 = (8.99×10^9)(5.0×10^−6)/(2.0^2) = 1.12×10^4 N/C. A positive source charge produces a field pointing away (+x); a negative source reverses direction (−x). Units N/C shown.

B. Mistake in magnitude (this equals kQ/r not kQ/r^2) and the statement about direction for Q = −5.0 μC is wrong: changing the sign of the source charge reverses the field, so it would not point the same way for both signs.

C. Direction error: the magnitude is correct but the student has reversed the directions. A positive source produces field away from the charge (+x here), not toward it; a negative source points toward the charge (−x).

D. Two errors: the magnitude 4.50×10^3 N/C is numerically wrong (likely a wrong r or factor), and the claim that a negative charge creates no defined field is false — negative charges produce fields with direction toward the charge; field exists and has units N/C.

Q20.

Answer: (a) Infinite plane (surface charge density sigma): Choose a cylindrical pillbox that intersects the plane, with flat faces of area A, one face on each side of the plane. By symmetry E is perpendicular to the plane and equal in magnitude on both sides.

Gauss's law: \Phi = \oint E·dA = E·A (top) + E·A (bottom) = 2EA = Q_enc/epsilon_0 = (sigma A)/epsilon_0.

Therefore E = sigma/(2 epsilon_0), directed normal to the plane, away from the plane if sigma>0.

(b) Infinite straight wire (linear charge density lambda): Choose a coaxial cylindrical Gaussian surface of radius r and length L. By symmetry E is radial and constant on the curved surface; flux through end caps is zero.

Gauss's law: \Phi = E (2 pi r L) = Q_enc/epsilon_0 = (lambda L)/epsilon_0.

Therefore E = lambda/(2 pi epsilon_0 r), directed radially outward if lambda>0.

(c) Thin spherical shell (radius R, total charge Q): Choose a spherical Gaussian surface of radius r centered on the shell.

- For r>R (outside): symmetry gives E radial and constant on sphere of area 4 pi r^2.

Gauss's law: E (4 pi r^2) = Q/epsilon_0 => E = Q/(4 pi epsilon_0 r^2), directed radially outward if Q>0. (Same as a point charge Q at center.)

- For r<R (inside): the Gaussian sphere encloses no charge (Q_enc = 0), so E (4 pi r^2) = 0 => E = 0 for r<R.

Working: (a) Infinite plane

Choose a thin cylindrical pillbox symmetric about the plane, face area A (m^2). Surface charge density sigma (C m^-2) means charge enclosed Q_enc = sigma A (C).

Flux: \Phi = E_top A + E_bottom A = 2 E A (E constant on faces, field parallel to area normals)

Gauss's law: 2 E A = Q_enc/epsilon_0 = (sigma A)/epsilon_0 => E = sigma/(2 epsilon_0).

Units: sigma in C m^-2, epsilon_0 in C^2 N^-1 m^-2 (or equivalently C^2/(N m^2)), so E in N C^-1 (V m^-1).

(b) Infinite straight wire

Choose a cylinder coaxial with the wire, radius r (m), length L (m). Linear charge density lambda (C m^-1) gives Q_enc = lambda L (C).

Flux: only curved surface contributes: \Phi = E (2 pi r L).

Gauss's law: E (2 pi r L) = Q_enc/epsilon_0 = (lambda L)/epsilon_0 => E = lambda/(2 pi epsilon_0 r).

Units: lambda in C m^-1, so E in N C^-1 as above.

(c) Thin spherical shell

Choose spherical Gaussian surface radius r.

If r > R: Q_enc = Q. Flux: E (4 pi r^2) = Q/epsilon_0 => E = Q/(4 pi epsilon_0 r^2). Units: Q (C), r (m).

If r < R: Q_enc = 0 (no charge inside shell). Flux: E (4 pi r^2) = 0 => E = 0.

These results assume electrostatic equilibrium, perfect symmetry, and that the plane/wire are effectively infinite so edge effects are negligible.
Rubric — Correct use of Gauss's law and symmetry to derive each electric field and state the Gaussian surface.
  Full credit (4 marks): All three parts have correct Gaussian surfaces stated and correct step-by-step application of Gauss's law leading to the correct final expressions: (a) E = sigma/(2 epsilon_0) normal to plane; (b) E = lambda/(2 pi epsilon_0 r) radial; (c) E = Q/(4 pi epsilon_0 r^2) for r>R and E=0 for r<R. Units or unit consistency shown in at least one part.
  Partial credit (2 marks): Work shows correct Gaussian surface and partially correct application for at least two parts or fully correct for one part and partially correct for another; final expressions contain at most one algebraic or symmetry error (e.g. missing factor of 2 or pi) and interior/exterior sphere cases correctly identified. Units may be omitted.
  No credit (0 marks): Incorrect or no use of Gauss's law and symmetry; Gaussian surfaces are inappropriate; final formulas are wrong for most parts or missing; sphere interior/exterior incorrectly treated.

Q21.

Correct: B. Rubbing transfers electrons (each with charge −e = −1.60×10^-19 C) from one rod to the other so each rod’s net charge is an integer multiple of e; the total charge of the two-rod system remains constant because no net charge is created or destroyed.

A. Incorrect: it asserts that rubbing creates new elementary charges and that total charge increases. This reflects the misconception that charge is produced by rubbing; it confuses transfer of charge with creation and violates charge conservation.
B. Correct: it states that rubbing transfers electrons (each −1.60×10^-19 C) so each object's net charge is an integer multiple of e and the total charge is conserved in the isolated two-rod system. This matches the microscopic origin

(electrons/protons) and quantisation of charge.

C. Incorrect: it claims fractional multiples of e can be transported, which contradicts quantisation. Elementary charges are indivisible (electrons/protons), so macroscopic charges must be integer multiples of e, not fractional parts like 0.5 e.

D. Incorrect: it says rubbing changes proton number (changing nuclei) and allows non-integer multiples of e; this is false for static charging—rubbing transfers electrons, not protons, and total charge is conserved. Changing protons would require nuclear reactions, not rubbing.

Q22.

Answer: (a) p = q(2a) = (2.0×10^−6 C)(2×0.050 m) = 2.0×10^−7 C·m. The torque magnitude is |τ| = p E sinθ. Here p E so sinθ = 1, therefore |τ| = (2.0×10^−7 C·m)(1.0×10^4 N·C^−1) = 2.0×10^−3 N·m. Using the right-hand rule for τ = p × E with p along +x and E along +y, τ points in the +z direction. So τ = (0, 0, 2.0×10^−3) N·m. (b) The net force on a dipole in a uniform electric field is zero because the forces on the two equal and opposite charges are equal in magnitude and opposite in direction and therefore cancel.

Working: Step 1: Compute dipole moment p.

p = q(2a) = (2.0×10^−6 C)(2×0.050 m) = (2.0×10^−6 C)(0.100 m) = 2.0×10^−7 C·m.

Step 2: Compute torque magnitude using |τ| = p E sinθ. The dipole moment (x-direction) is perpendicular to E (y-direction), so θ = 90°, sinθ = 1.

|τ| = p E = (2.0×10^−7 C·m)(1.0×10^4 N·C^−1) = 2.0×10^−3 N·m.

Step 3: Determine vector direction via cross product τ = p × E. In components p = (p,0,0), E = (0,E,0) => p×E = (0,0,pE). Thus τ = (0, 0, 2.0×10^−3) N·m, i.e. along +z.

Step 4: Net force on dipole in uniform field: Forces on +q and −q are +qE and −qE; they are equal and opposite, so they sum to zero. Therefore net force = 0 N.

Rubric — Correct calculation of torque (magnitude with units and vector direction) and correct statement about net force with brief reason.

Full credit (4 marks): Correct numerical value of p and torque magnitude with units (2.0×10^−3 N·m), correct vector direction (+z or (0,0,2.0×10^−3) N·m) and clear statement that net force is zero with brief correct reason (forces on charges cancel in a uniform field).

Partial credit (2 marks): Correct torque magnitude (or correct p) but either units missing or vector direction omitted/incorrect, or correct torque but net force statement missing or incorrect; reasoning partially correct.

No credit (0 marks): Numerical value and units for torque incorrect and no correct statement that net force is zero (or incorrect reasoning).

Q23.

Answer: Electric field at midpoint: E_total = 1.44 × 10^6 N/C, directed to the right (toward the −2.0 μC). Force on +1.0 μC test charge: F = 1.44 N, directed to the right.

Working: Distance from each charge to the midpoint: r = 0.25 m.

Electric field magnitude from the +8.0 μC charge at the midpoint:

E_+ = k |q_+| / r^2 = (8.99×10^9 N·m^2/C^2) × (8.0×10^−6 C) / (0.25 m)^2.

(0.25 m)^2 = 0.0625 m^2, so

E_+ = 8.99×10^9 × 8.0×10^−6 / 0.0625 = 8.99×10^9 × 1.28×10^−4 = 1.15072×10^6 N/C.

Direction: field from a positive charge points away from the +8.0 μC; at the midpoint this is to the right.

Electric field magnitude from the −2.0 μC charge at the midpoint:

E_- = k |q_-| / r^2 = (8.99×10^9) × (2.0×10^−6) / 0.0625 = 8.99×10^9 × 3.2×10^−5 = 2.8768×10^5 N/C.

Direction: field points toward the negative charge; at the midpoint (left of the −2.0 μC) this is also to the right.

Because both fields point to the right, they add:

E_total = E_+ + E_- = 1.15072×10^6 + 2.8768×10^5 = 1.4384×10^6 N/C 1.44×10^6 N/C to the right.

Force on the test charge q_test = +1.0 μC = 1.0×10^−6 C:

F = q_test × E_total = (1.0×10^−6 C) × (1.4384×10^6 N/C) = 1.4384 N 1.44 N to the right.

(Units carried through: N/C × C = N.)

Rubric — Calculation of electric field at midpoint and resulting force on a test charge, with correct magnitudes, directions, units, and clear method.

Full credit (4 marks): Correct calculation of E from each charge (or directly), correct sum E_total = 1.44×10^6 N/C with direction to the right, and correct force F = 1.44 N to the right. All units shown and method clearly presented.

Partial credit (2 marks): Method is correct and shown (use of E = kq/r^2 and superposition) and at least one correct field magnitude computed, but arithmetic or rounding error leads to a numerical answer within ~20% of correct values, or direction omitted/partly incorrect. Force may be proportional to E_total but numerically inconsistent by the same factor. Units present.

Minimal credit (0 marks): Only a brief or incorrect attempt: no clear use of kq/r^2 and superposition, or numbers/units missing, or final answers incorrect with no correct intermediate values. Shows little understanding of fields adding and relation F = qE.

Q24.

Answer: (a) Voltage for 1.0 cm gap: 3.0 × 10^6 V/m × 0.010 m = 3.0 × 10^4 V = 30 kV.

(b) Voltage for 2.0 mm gap: 3.0 × 10^6 V/m × 0.0020 m = 6.0 × 10^3 V = 6.0 kV.

(c) The 1.0 cm spark requires the larger voltage (30 kV > 6.0 kV). Everyday reason: small shocks from handles are more common because they need a much smaller gap/voltage to discharge; also people and objects are often close so the required field is reached at lower voltages, whereas 1 cm sparks require larger charge buildup which happens less often.

Working: (a) Convert gap to metres: 1.0 cm = 0.010 m.

Use V = E × d where E = 3.0 × 10^6 V/m.

V = (3.0 × 10^6 V/m) × (0.010 m) = 3.0 × 10^4 V = 30 000 V = 30 kV.

(b) Convert gap to metres: 2.0 mm = 0.0020 m.

V = (3.0 × 10^6 V/m) × (0.0020 m) = 6.0 × 10^3 V = 6 000 V = 6.0 kV.

(c) Compare: 30 kV (1.0 cm) is greater than 6.0 kV (2.0 mm), so the 1.0 cm spark requires the larger voltage. Everyday reason: smaller gaps need less charge/voltage to reach breakdown, so small shocks occur more often; building up 30 kV for a 1 cm spark is less common than the few kilovolts needed for a tiny spark.

Rubric — Correct calculation of voltages for both gap lengths and a correct comparison with an everyday reason.

Full marks (4 marks): Correct numerical answers with units: 30 kV for 1.0 cm and 6.0 kV for 2.0 mm, clear statement that 1.0 cm requires the larger voltage, and a plausible everyday reason explaining why small-handle shocks are more common.

Partial credit (2 marks): One voltage correct with units and the other roughly correct (within a factor of 2) or both numerical answers given but one or both lack units; comparison stated correctly but everyday reason is missing or weak.

No credit (0 marks): No correct numerical calculation (answers missing or wrong by more than a factor of 2), and no correct comparison or reason.